Prodigy Anesthesia Quick Review

Airway Management

What is denitrogenation, how is it performed, and what are the signs that it has been performed correctly?	Proper denitrogenation should be performed using a high flow of oxygen with a fully-open pop-off valve, a leak-free mask fit, and the allowance of tidal breathing for 2-3 minutes or a series of 4 vital capacity breaths. Signs that denitrogenation is being accomplished include: noticing the reservoir bag emptying and refilling, end-tidal CO2 approaches 40 mmHg, and end-tidal O2 approaches 85%. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 551.*
Are the relationships between the risk of airway difficulty and Mallampati scores, body mass index, tongue size, and thyromental distance considered direct or indirect relationships?	Airway management risk increases as the Mallampati score, body mass index, and size of the patient's tongue increase (a direct relationship). The risk also increases as the thyromental distance decreases (inverse relationship). *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 431-433.*
Tonic activity in which muscles is responsible for preventing the soft palate from falling back against the posterior pharynx and occluding the airway?	The tensor palati, levator palati, palatopharyngeus, and palatoglossus keep the soft palate from falling back against the posterior pharynx. The genioglossus prevents the tongue from falling back against the posterior pharynx. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 489.*
The tongue falls back against the posterior pharynx resulting in airway obstruction when what muscle relaxes?	Loss of motor tone to the genioglossus allows the tongue to fall back against the posterior pharynx resulting in airway obstruction. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 314.*
What are three factors that can reduce the incidence of post-operative hoarseness due to intubation?	The use of a smaller endotracheal tube, and using low-pressure cuffs will decrease the incidence of postoperative hoarseness due to intubation. Studies have demonstrated that the use of a water-soluble lubricant or gel containing local anesthetic solution applied to the tip or cuff of the ETT does not decrease the incidence of sore throat or hoarseness. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 335.*
What is a probable contraindication to nasotracheal intubation?	Nasotracheal intubation may be contraindicated in a patient with severe mid-facial trauma (such as a maxillary fracture) which may represent an existing cribriform plate fracture. If the cribriform plate is unstable, the tracheal tube could potentially pass through it into the cranial vault. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 330.*

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What are the contraindications to blind nasal intubation? In what circumstances would it be less likely to be successful? Is general anesthesia required for this procedure?	Contraindications to blind nasal intubation include nasal pathology, coagulopathy, thrombocytopenia, severe midface trauma, and previous transphenoidal surgery (the latter two of which could permit intracranial intubation). Blind nasal intubation is not contraindicated but likely to be less successful in patients with laryngeal masses, edema, or scarring from a previous surgery. Blind nasal intubation is possible in an awake patient, and can even be easier as a patient encouraged to breathe rapidly may facilitate introduction of the endotracheal tube through the larynx. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 570.*
What are the indications for a cricothyrotomy?	Indications for a cricothyrotomy include: cannot intubate-cannot ventilate scenarios, traumatic head, neck or facial injuries that make intubation difficult or impossible, upper airway obstruction, or definitive airway management when intubation is not possible. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 459.*
How is a cricothyrotomy performed?	A cricothyrotomy is performed by inserting a large-bore intravenous catheter or cannula into the cricothyroid membrane which lies between the inferior border of the thyroid cartilage and the superior border of the cricoid cartilage. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 332-333.*
What are the primary advantages of a cricothyrotomy over a surgical tracheostomy during emergency airway management?	A cricothyrotomy can result in patient ventilation in about 1-2 minutes and can be performed by a skilled anesthesia provider. A surgical airway requires a skilled surgeon and can take up to 30 minutes to perform. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 459-461.*
Where on the neck is a tracheotomy usually performed?	It is typically performed at the level of the fourth to sixth tracheal ring. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 461.*
Can an anesthetist perform an emergency surgical tracheotomy?	No, a tracheotomy is not a procedure to be performed by an anesthesia provider. It requires a surgically-trained professional. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 460.*
What are the factors that are associated with the likelihood of a difficult mask ventilation?	male gender, age greater than 57 years, increased Mallampati score, short thyromental distance, elevated BMI, large tongue, presence of a beard, sleep apnea, anatomical neck abnormalities, snoring, and a limited ability to protrude the jaw. *Dorsch JA, Dorsch SE. A Practical Approach to Anesthesia Equipment. Philadelphia, PA: Lippincott Williams and Wilkins, 2011: 249.*

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What are the effects of unilateral paralysis of the recurrent laryngeal nerve? Bilateral paralysis of the recurrent laryngeal nerve?

Unilateral paralysis of the recurrent laryngeal nerve can result in changes in phonation and hoarseness. Stridor and acute airway compromise can result from bilateral paralysis of the recurrent laryngeal nerve.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 312.

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